Survey (Set 3)

When developers add new test cases into an existing test suite, do the developers look at the existing test cases and then reuse different attributes of existing test cases (such as variable naming, comments, test case naming, set of source code methods, overall test case organization/structure etc.) to derive the new testcases?

In order to answer the above question, we have designed this survey. The survey shows 20 pairs of unit test cases collected from open source projects. For each pair the participant is requested to indicate to what extent a developer *may/may not* have used a specific attribute to write the new test case.

Please answer all the following questions to complete the study. Thanks in advance.

* Indicates required question

Example 1: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
public void testWrite_charArrayToOutputStream_nullData() throws Exception {
  ByteArrayOutputStream baout = new ByteArrayOutputStream();
  YellOnFlushAndCloseOutputStream out = new YellOnFlushAndCloseOutputStream(baout, true, true);
  IOUtils.write((char[]) null, out);
  out.off();
  out.flush();
  assertEquals("Sizes differ", 0, baout.size());
}
public void testWrite_charSequenceToOutputStream_nullData() throws Exception {
  ByteArrayOutputStream baout = new ByteArrayOutputStream();
  YellOnFlushAndCloseOutputStream out = new YellOnFlushAndCloseOutputStream(baout, true, true);
  IOUtils.write((CharSequence) null, out);
  out.off();
  out.flush();
  assertEquals("Sizes differ", 0, baout.size());
}
```

1. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

2. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

3. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

4. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

5. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

6. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

7. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

8. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

9. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 2: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void verifyChecksum() throws IOException {
    final XXHash32 h = new XXHash32();
    try (final FileInputStream s = new FileInputStream(file)) {
        final byte[] b = toByteArray(s);
        h.update(b, 0, b.length);
    }
    Assert.assertEquals(""checksum for "" + file.getName(),
        expectedChecksum, Long.toHexString(h.getValue()));
}
@Test
public void verifyIncrementalChecksum() throws IOException {
    final XXHash32 h = new XXHash32();
    try (final FileInputStream s = new FileInputStream(file)) {
        final byte[] b = toByteArray(s);
        // Hit the case where the hash should be reset
        h.update(b[0]);
        h.reset();
        // Pass in chunks
        h.update(b[0]);
        h.update(b, 1, b.length - 2);
        h.update(b, b.length - 1, 1);
        // Check the hash ignores negative length
        h.update(b, 0, -1);
    }
    Assert.assertEquals(""checksum for "" + file.getName(),
        expectedChecksum, Long.toHexString(h.getValue()));
}
```

10. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

11. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

12. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

13. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

14. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

15. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

16. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

17. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

18. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 3: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"@Test
public void testAppendStringBuilderNull() {
    final TextStringBuilder sb1 = new TextStringBuilder();
    final StringBuilder b = null;
    assertEquals("", sb1.append(b).toString());
    final TextStringBuilder sb2 = new TextStringBuilder();
    assertEquals("", sb2.append(b, 0, 0).toString());
}
@Test
public void testDrainChar() {
    final String str = ""abc"";
    //
    final TextStringBuilder sb1 = new TextStringBuilder(str);
    assertEquals('a', sb1.drainChar(0));
    assertEquals(""bc"", sb1.toString());
    //
    final TextStringBuilder sb2 = new TextStringBuilder(str);
    assertEquals('c', sb2.drainChar(str.length() - 1));
    assertEquals(""ab"", sb2.toString());
    //
    final TextStringBuilder sb3 = new TextStringBuilder(str);
    assertThrows(IndexOutOfBoundsException.class, () -> sb3.drainChar(str.length()));
    assertThrows(IndexOutOfBoundsException.class, () -> sb3.drainChar(1000));
}"
```

19. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

20. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

21. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

22. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

23. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

24. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

25. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

26. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

27. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 4: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
public void testUTF16WithSingleByteWrite() throws IOException {
    testWithSingleByteWrite(TEST_STRING, "UTF-16");
}
public void testUTF16BEWithSingleByteWrite() throws IOException {
    testWithSingleByteWrite(TEST_STRING, "UTF-16BE");
}
```

28. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

29. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

30. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

31. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

32. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

33. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

34. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

35. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

36. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 5: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testAvailableAfterClose() throws IOException {
    try (RandomAccessFileInputStream inputStream =
        new RandomAccessFileInputStream(createRandomAccessFile(), true)) {
        inputStream.close();
        assertEquals(0, inputStream.available());
    }
}
@Test
public void testReadAfterClose() throws IOException {
    try (RandomAccessFileInputStream inputStream =
         new RandomAccessFileInputStream(createRandomAccessFile(), true)) {
        inputStream.close();
        assertEquals(IOUtils.EOF, inputStream.read());
    }
}
```

37. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

38. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

39. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

40. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

41. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

42. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

43. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

44. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

45. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 6: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testDeleteCharAt() {
    final String str = "abc";
    //
    final TextStringBuilder sb1 = new TextStringBuilder(str);
    sb1.deleteCharAt(0);
    assertEquals("bc", sb1.toString());
    //
    final TextStringBuilder sb2 = new TextStringBuilder(str);
    sb2.deleteCharAt(str.length() - 1);
    assertEquals("ab", sb2.toString());
    //
    final TextStringBuilder sb3 = new TextStringBuilder(str);
    assertThrows(IndexOutOfBoundsException.class, () -> sb3.deleteCharAt(str.length()));
    assertThrows(IndexOutOfBoundsException.class, () -> sb3.deleteCharAt(1000));
}
@Test
public void testDrainChar() {
    final String str = "abc";
    //
    final TextStringBuilder sb1 = new TextStringBuilder(str);
    assertEquals('a', sb1.drainChar(0));
    assertEquals("bc", sb1.toString());
    //
    final TextStringBuilder sb2 = new TextStringBuilder(str);
    assertEquals('c', sb2.drainChar(str.length() - 1));
    assertEquals("ab", sb2.toString());
    //
    final TextStringBuilder sb3 = new TextStringBuilder(str);
    assertThrows(IndexOutOfBoundsException.class, () -> sb3.drainChar(str.length()));
    assertThrows(IndexOutOfBoundsException.class, () -> sb3.drainChar(1000));
}
```

46. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

47. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

48. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

49. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

50. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

51. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

52. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

53. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

54. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 7: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void simpleBehaviourTest() {
    String h = "<div><p class=foo><a href='http://evil.com'>Hello <b id=bar>there</b>!</a></div>";
    String cleanHtml = Jsoup.clean(h, Whitelist.simpleText());
    assertEquals("Hello <b>there</b>!", cleanHtml);
}
@Test
public void simpleBehaviourTest2() {
    String h = "Hello <b>there</b>!";
    String cleanHtml = Jsoup.clean(h, Whitelist.simpleText());
    assertEquals("Hello <b>there</b>!", cleanHtml);
}
```

55. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

56. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

57. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

58. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

59. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

60. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

61. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

62. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

63. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 8: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
@Test
public void testReadLines_Reader() throws Exception {
    final File file = TestUtils.newFile(temporaryFolder, ""lines.txt"");
    Reader in = null;
    try {
        final String[] data = { "hello", "/u1234", "", "this is", "some text" };
        TestUtils.createLineBasedFile(file, data);
        in = new InputStreamReader(Files.newInputStream(file.toPath()));
        final List<String> lines = IOUtils.readLines(in);
        assertEquals(Arrays.asList(data), lines);
        assertEquals(-1, in.read());
    } finally {
        IOUtils.closeQuietly(in);
        TestUtils.deleteFile(file);
    }
}
@Test
public void testReadLines_CharSequence() throws IOException {
    final File file = TestUtils.newFile(temporaryFolder, ""lines.txt"");
    CharSequence csq = null;
    try {
        final String[] data = { "hello", "/u1234", "", "this is", "some text" };
        TestUtils.createLineBasedFile(file, data);
        csq = new String(Files.readAllBytes(file.toPath()), StandardCharsets.UTF_8);
        final List<String> lines = IOUtils.readLines(csq);
        assertEquals(Arrays.asList(data), lines);
    } finally {
        TestUtils.deleteFile(file);
    }
}
```

64. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

65. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

66. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

67. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

68. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

69. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

70. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

71. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

72. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 9: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testAppendToWriter() throws Exception {
    final TextStringBuilder sb = new TextStringBuilder(""1234567890"");
    final StringWriter writer = new StringWriter();
    writer.append(""Test "");
    sb.appendTo(writer);
    assertEquals(""Test 1234567890"", writer.toString());
}
@Test
public void testAppendToAppendable() throws Exception {
    final TextStringBuilder sb = new TextStringBuilder(""1234567890"");
    final StringWriter writer = new StringWriter();
    writer.append(""Test"");
    // TODO Use Commons IO NullApendable.INSTANCE
    sb.appendTo(new Appendable() {

        @Override
        public Appendable append(final char c) throws IOException {
            return this;
        }

        @Override
        public Appendable append(final CharSequence csq) throws IOException {
            return this;
        }

        @Override
        public Appendable append(final CharSequence csq,
                                 final int start, final int end) throws IOException {
            return this;
        }
    });
    assertEquals(""Test"", writer.toString());
}
```

73. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

74. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

75. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

76. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

77. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

78. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

79. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

80. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

81. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 10: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testDecodeHexStringOddCharacters() {
    try {
        new Hex().decode("6");
        fail("An exception wasn't thrown when trying to decode an odd number of
            characters");
    } catch (final DecoderException e) {
        // Expected exception
    }
}
@Test
public void testDecodeByteArrayOddCharacters() {
    try {
        new Hex().decode(new byte[] { 65 });
        fail("An exception wasn't thrown when trying to decode an odd number of
            characters");
    } catch (final DecoderException e) {
        // Expected exception
    }
}
```

82. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

83. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

84. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

85. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

86. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

87. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

88. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

89. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

90. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 11: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testBytesSkipped() throws IOException {
    for (final InputStream inputStream : inputStreams) {
        assertEquals(1024, inputStream.skip(1024));
        for (int i = 1024; i < expectedRandomBytes.length; i++) {
            assertEquals(expectedRandomBytes[i], (byte) inputStream.read());
        }
    }
}
@Test
public void testReadOneByOne() throws IOException {
    for (final InputStream inputStream : inputStreams) {
        for (final byte randomByte : expectedRandomBytes) {
            assertEquals(randomByte, (byte) inputStream.read());
        }
    }
}
```

91. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

92. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

93. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

94. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

95. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

96. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

97. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

98. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

99. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 12: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testJdbcPrinter() throws IOException, ClassNotFoundException, SQLException {
    final StringWriter sw = new StringWriter();
    try (final Connection connection = getH2Connection()) {
        setUpTable(connection);
        try (final Statement stmt = connection.createStatement();
            final CSVPrinter printer = new CSVPrinter(sw, CSVFormat.DEFAULT);
            final ResultSet resultSet = stmt.executeQuery("select ID, NAME, TEXT from TEST")) {
            printer.printRecords(resultSet);
        }
    }
    assertEquals("1,r1,\"long text 1\"" + recordSeparator
                    + "2,r2,\"" + longText2 + "\"" + recordSeparator, sw.toString());
}
@Test
public void testJdbcPrinterWithFirstEmptyValue2() throws IOException, ClassNotFoundException,
    SQLException {
    final StringWriter sw = new StringWriter();
    try (final Connection connection = getH2Connection()) {
        try (final Statement stmt = connection.createStatement();
            final ResultSet resultSet = stmt.executeQuery("select '' AS EMPTYVALUE from DUAL");
            final CSVPrinter printer = CSVFormat.DEFAULT.withHeader(resultSet).print(sw)) {
            printer.printRecords(resultSet);
        }
    }
    assertEquals("EMPTYVALUE" + recordSeparator +
                    "\"\"" + recordSeparator, sw.toString());
}
```

100. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

101. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

102. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

103. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

104. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

105. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

106. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

107. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

108. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 13: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testAaclan() throws EncoderException {
    this.checkEncoding("0856", "Aaclan");
}
@Test(expected = org.junit.ComparisonFailure.class)
public // Ensure that override still allows tests to work
void testCanFail() throws EncoderException {
    this.checkEncoding("/", "Fehler");
}
```

109. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

110. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

111. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

112. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

113. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

114. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

115. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

116. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

117. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 14: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void testEmptyLineBehaviourCSV() throws Exception {
    String[] codes = { "hello,\r\n\r\n\r\n", "hello,\n\n\n",
    "hello,\"\"\r\n\r\n\r\n", "hello,\"\"\n\n\n" };
    String[][] res = { // CSV format ignores empty lines
    { "hello", "" } };
    String code;
    for (int codeIndex = 0; codeIndex < codes.length; codeIndex++) {
        code = codes[codeIndex];
        CSVParser parser = new CSVParser(new StringReader(code));
        String[][] tmp = parser.getAllValues();
        assertEquals(res.length, tmp.length);
        assertTrue(tmp.length > 0);
        for (int i = 0; i < res.length; i++) {
            assertTrue(Arrays.equals(res[i], tmp[i]));
        }
    }
}
public void testExcelFormat1() throws IOException {
    String code = "value1,value2,value3,value4\r\na,b,c,d\r\n  x,,," +
    "\r\n\r\n\"\"\"hello\"\"\",\"  \"\"world\"\"\",\"abc\ndef\",\r\n";
    String[][] res = { { "value1", "value2", "value3", "value4" },
    { "a", "b", "c", "d" }, { "  x", "", "", "" }, { "" },
    { "\"hello\"", "  \"world\"", "abc\ndef", "" } };
    CSVParser parser = new CSVParser(new StringReader(code), CSVFormat.EXCEL);
    String[][] tmp = parser.getAllValues();
    assertEquals(res.length, tmp.length);
    assertTrue(tmp.length > 0);
    for (int i = 0; i < res.length; i++) {
        assertTrue(Arrays.equals(res[i], tmp[i]));
    }
}
```

118. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

119. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

120. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

121. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

122. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

123. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

124. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

125. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

126. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 15: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testToMapWithNoHeader() throws Exception {
    try (final CSVParser parser = CSVParser.parse(""a,b"", CSVFormat.newFormat('
        ,'))) {
        final CSVRecord shortRec = parser.iterator().next();
        final Map<String, String> map = shortRec.toMap();
        assertNotNull(map, ""Map is not null."");
        assertTrue(map.isEmpty(), ""Map is empty."");
    }
}
@Test
public void testDuplicateHeaderGet() throws IOException {
    final String csv = ""A,A,B,B\n1,2,5,6\n"";
    final CSVFormat format = CSVFormat.DEFAULT.builder().setHeader().build();
    try (final CSVParser parser = CSVParser.parse(csv, format)) {
        final CSVRecord record = parser.nextRecord();
        assertAll(""Test that it gets the last instance of a column when there
            are duplicate headings"", () -> assertEquals(""2"", record.get(""A""
            )), () -> assertEquals(""6"", record.get(""B""")));
    }
}
```

127. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

128. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

129. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

130. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

131. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

132. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

133. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

134. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

135. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 16: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
"/**
 * Test the iterator using only the nextLine() method.
 */
public void testNextLineOnlyDefaultEncoding() throws Exception {
    File testFile = new File(getTestDirectory(), ""LineIterator-nextOnly.txt"");
    List<String> lines = createLinesFile(testFile, 3);
    LineIterator iterator = FileUtils.lineIterator(testFile);
    assertLines(lines, iterator);
}
/**
 * Test the iterator using only the nextLine() method.
 */
public void testNextLineOnlyNullEncoding() throws Exception {
    String encoding = null;
    File testFile = new File(getTestDirectory(), ""LineIterator-nextOnly.txt"");
    List<String> lines = createLinesFile(testFile, encoding, 3);
    LineIterator iterator = FileUtils.lineIterator(testFile, encoding);
    assertLines(lines, iterator);
}"
```

136. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

137. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

138. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

139. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

140. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

141. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

142. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

143. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

144. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 17: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testParsePathCharsetNullFormat() throws IOException {
    final Path path =
            Paths.get("src/test/resources/org/apache/commons/csv/CSVFileParser/test.csv");
    try (CSVParser parser = CSVParser.parse(path, Charset.defaultCharset(), null)) {
        // null maps to DEFAULT.
        parseFully(parser);
    }
}
@Test
public void testParseInputStreamCharsetNullFormat() throws IOException {
    try (InputStream in = Files.newInputStream(
            Paths.get("src/test/resources/org/apache/commons/csv/CSVFileParser/test.csv"));
        CSVParser parser = CSVParser.parse(in, Charset.defaultCharset(), null)) {
        // null maps to DEFAULT.
        parseFully(parser);
    }
}
```

145. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

146. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

147. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

148. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

149. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

150. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

151. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

152. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

153. Please indicate if any other attributes were used (not listed in above)

_____

Untitled Section

Example 18: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```java
public void testSkipReader() throws Exception {
    final int size = 1027;
    Reader input = new CharArrayReader(new char[size]);
    IOUtils.skipFully(input, 0);
    IOUtils.skipFully(input, size - 3);
    try {
        IOUtils.skipFully(input, -1);
        fail(""Should have failed with IllegalArgumentException"");
    } catch (IllegalArgumentException expected) {
        // expected
    }
    try {
        IOUtils.skipFully(input, 5);
        fail(""Should have failed with IOException"");
    } catch (IOException expected) {
        // expected
    }
    IOUtils.closeQuietly(input);
}
public void testReadStream() throws Exception {
    final int size = 1027;
    byte[] buffer = new byte[size];
    InputStream input = new ByteArrayInputStream(new byte[size]);
    try {
        IOUtils.readFully(input, buffer, 0, -1);
        fail(""Should have failed with IllegalArgumentException"");
    } catch (IllegalArgumentException expected) {
        // expected
    }
    IOUtils.readFully(input, buffer, 0, 0);
    IOUtils.readFully(input, buffer, 0, size - 1);
    try {
        IOUtils.readFully(input, buffer, 0, 2);
        fail(""Should have failed with EOFxception"");
    } catch (EOFException expected) {
        // expected
    }
    IOUtils.closeQuietly(input);
}
```

154. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

155. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

156. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

157. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

158. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

159. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

160. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

161. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

162. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 19: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
// encapsulator tokenizer (multi line, delimiter in string)
@Test
public void testNextToken5() throws IOException {
    final String code = "a,\""foo\n\"",b\n\""foo\n  baar ,,,\""\n\""\n\t \n\"";
    try (final Lexer parser = createLexer(code, CSVFormat.DEFAULT)) {
        assertThat(parser.nextToken(new Token()), matches(TOKEN, ""a""));
        assertThat(parser.nextToken(new Token()), matches(TOKEN, ""foo\n""));
        assertThat(parser.nextToken(new Token()), matches(EORECORD, ""b""));
        assertThat(parser.nextToken(new Token()), matches(EORECORD, "foo\n  baar ,,,"));
        assertThat(parser.nextToken(new Token()), matches(EOF, ""\n\t \n""));
    }
}
@Test
public void testEOFWithoutClosingQuote() throws Exception {
    final String code = ""a,\""b"";
    try (final Lexer parser = createLexer(code,
            CSVFormat.Builder.create().setLenientEof(true).build())) {
        assertThat(parser.nextToken(new Token()), matches(TOKEN, ""a""));
        assertThat(parser.nextToken(new Token()), matches(EOF, ""b""));
    }
    try (final Lexer parser = createLexer(code,
            CSVFormat.Builder.create().setLenientEof(false).build())) {
        assertThat(parser.nextToken(new Token()), matches(TOKEN, ""a""));
        assertThrows(IOException.class, () -> parser.nextToken(new Token()));
    }
}
```

163. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

164. To what extent the developer may have used the **"Variable Naming Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

165. To what extent the developer may have used the **"Variable or Object Construction**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

166. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

167. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

168. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

169. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

170. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

171. Please indicate if any other attributes were used (not listed in above)

Untitled Section

Example 20: Following is a pair of test cases where the first one is an existing test case and the later one is the new test case. Please indicate to what extent a developer may have used the following attributes while writing the new test cases where -
(5=High extent of use)
(1=Low extent of use)

```
@Test
public void testSetLength() {
    final int length = 99;
    final RandomStringGenerator generator =
            new RandomStringGenerator.Builder().build();
    final String str = generator.generate(length);
    assertThat(codePointLength(str)).isEqualTo(length);
}
@Test
public void testSelectFromNullCharVarargs() {
    final int length = 5;
    final RandomStringGenerator generator =
            new RandomStringGenerator.Builder().selectFrom(null).build();
    final String randomText = generator.generate(length);
    assertThat(codePointLength(randomText)).isEqualTo(length);
}
```

172. To what extent the developer may have used the "**Testcase Naming Style**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

173. To what extent the developer may have used the **"Variable Naming Style"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

174. To what extent the developer may have used the **"Variable or Object Construction"** (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

175. To what extent the developer may have used the "**Comments**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

176. To what extent the developer may have used the "**Set of Source Code Methods Invoked**" *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

177. To what extent the developer may have used the "**Assertions (i.e. the attributes/methods being invoked on the asserted object**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

178. To what extent the developer may have used the "**Testcase Structure/Organization Style**" (if any) *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

179. Overall did the developer used the existing test case to write the new one? *

Mark only one oval.

1	2	3	4	5
◯	◯	◯	◯	◯

180. Please indicate if any other attributes were used (not listed in above)

_____

Demographic Information

181. What is you current post * Dropdown

Mark only one oval.

- PhD Student
- Masters Student
- Undergraduate Student
- Software Developer
- Others

182. How many years of programming experience you have? Dropdown

Mark only one oval.

- 0-1 year
- 1-3 years
- 3-5 years
- more than 5 years

183. Do you have experience in writing unit test cases?

Mark only one oval.

- Yes
- No
- Maybe

This content is neither created nor endorsed by Google.

Google Forms